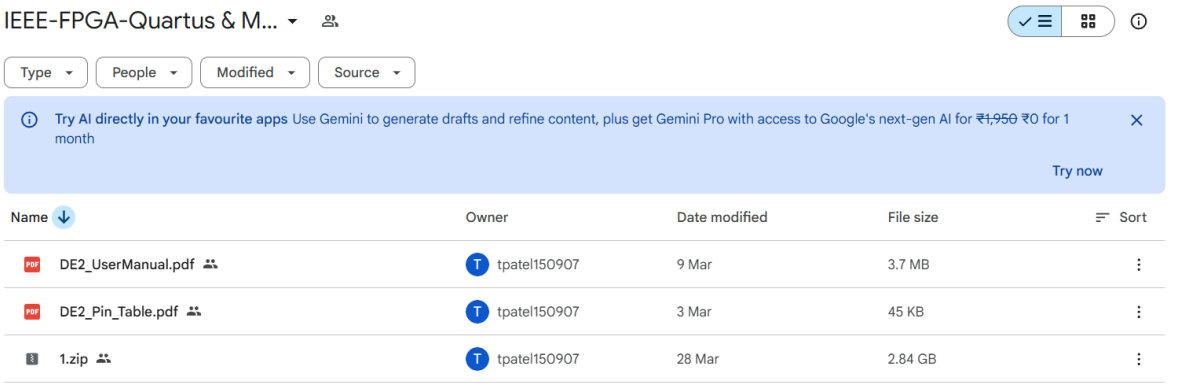
1. Downloading the Files

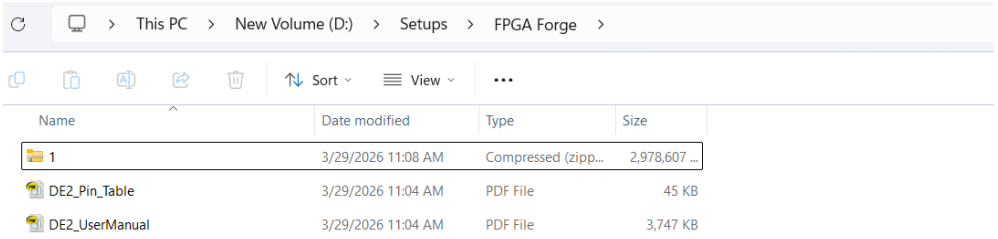
Since the Altera DE2 uses the **Cyclone II** chip, modern versions of Quartus will not work. We are using **Quartus II 13.0 SP1 (Web Edition)**.

1. Open the Google Drive link provided by the IEEE team.

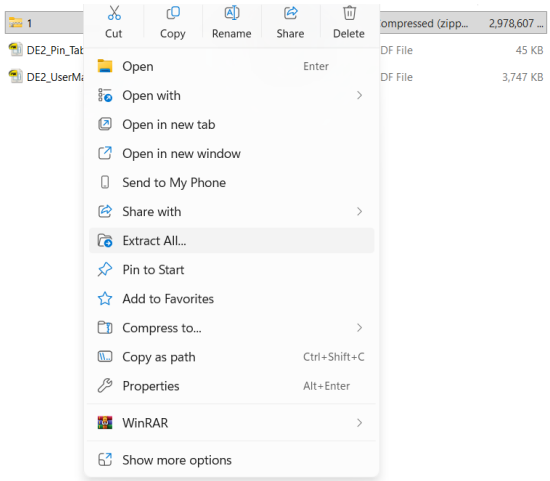


2. **Download Everything:** * **1.zip (2.84 GB):** This contains the Quartus II software, Cyclone 2 and ModelSim.
   a. **DE2_UserManual.pdf:** Keep this for hardware reference.
   b. **DE2_Pin_Table.pdf:** You will need this later to find which pins connect to your Servo motors.

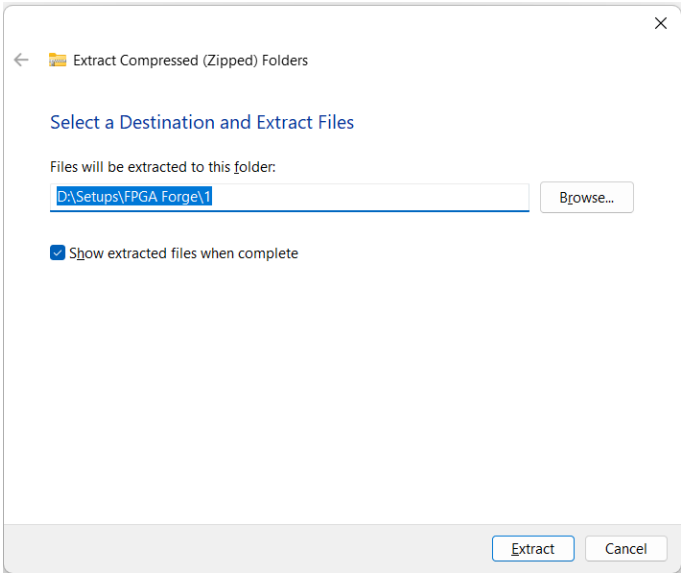
   [Keep all downloaded files in the same folder.]



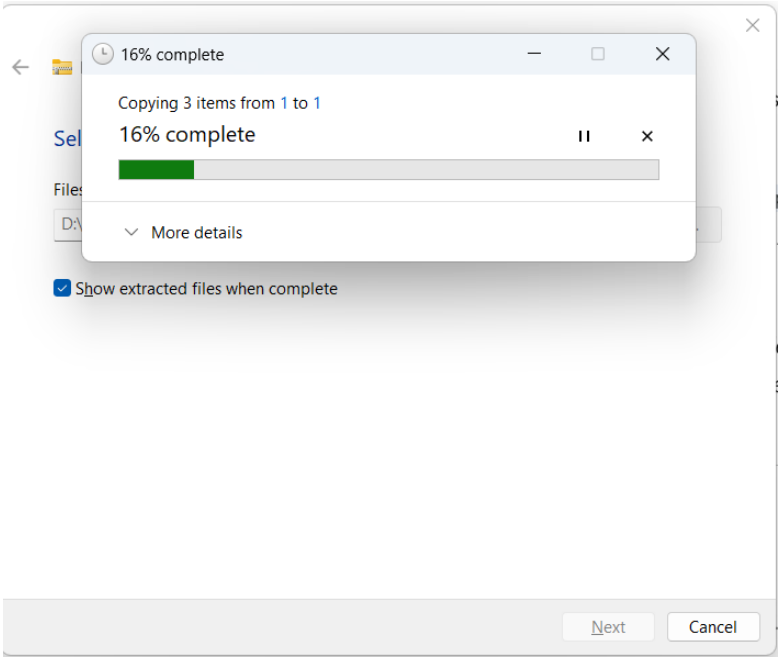
3. **Unzip the Software:** * Locate 1.zip in your Downloads folder.
   a. **Right-click** it and select **"Extract All..."**.



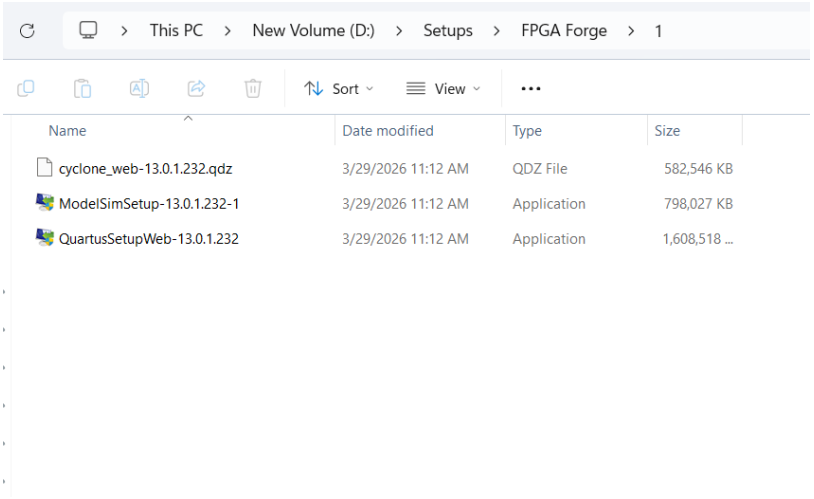
b. Choose a simple folder path



c. **Wait** for the extraction to finish

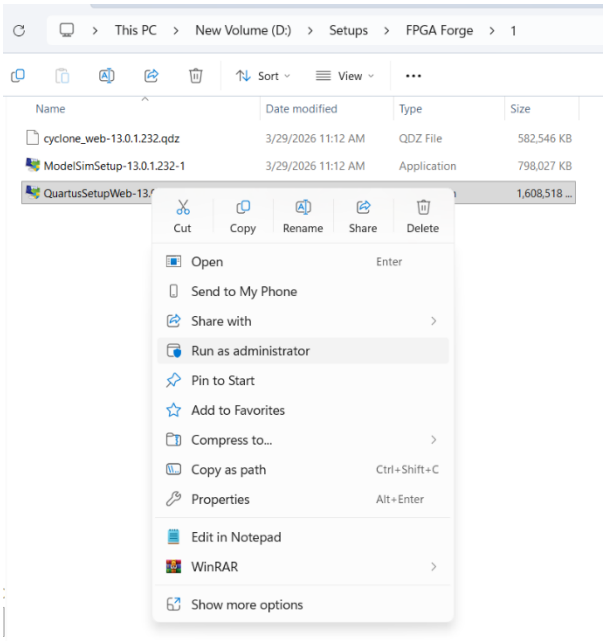


d. You should now see the .exe installers inside that folder.

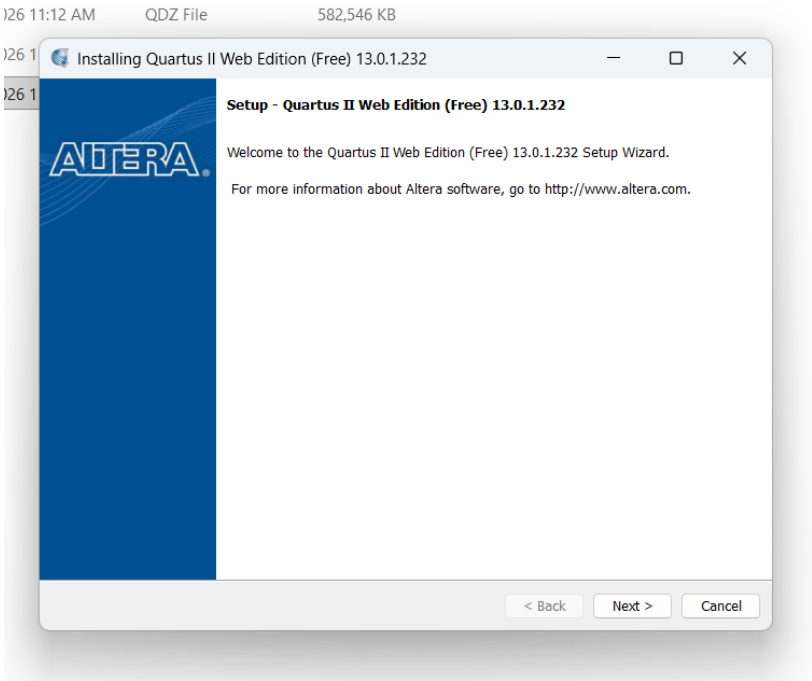


2. Installing Quartus II

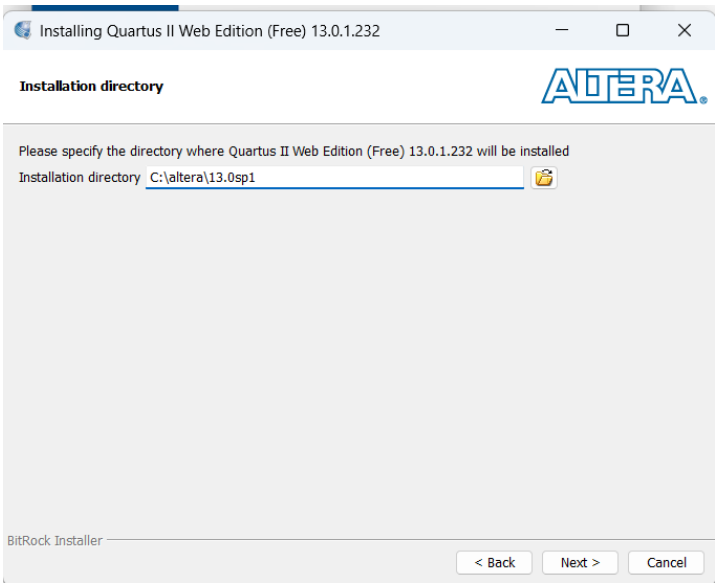
1. Navigate into the folder where you extracted **1.zip**.
2. **Run the Installer:** Right-click QuartusSetup-13.0.1.232.exe and select **Run as Administrator**.



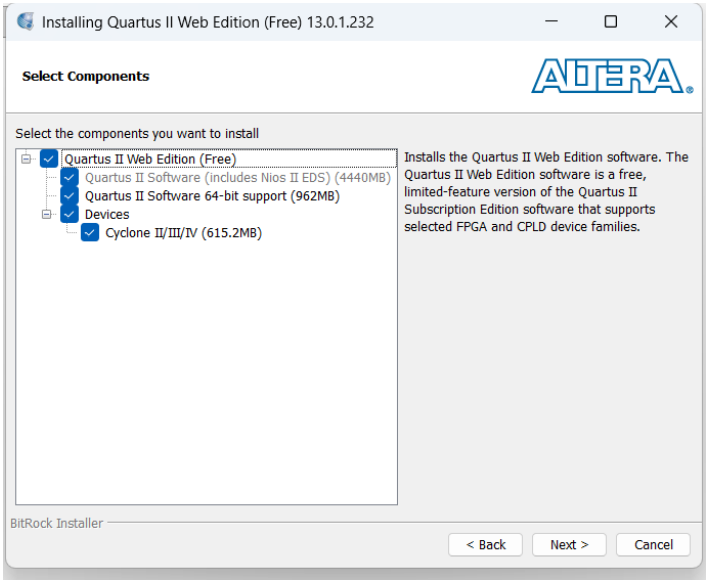
3. **Introduction:** Click **Next**. Accept the License Agreement.



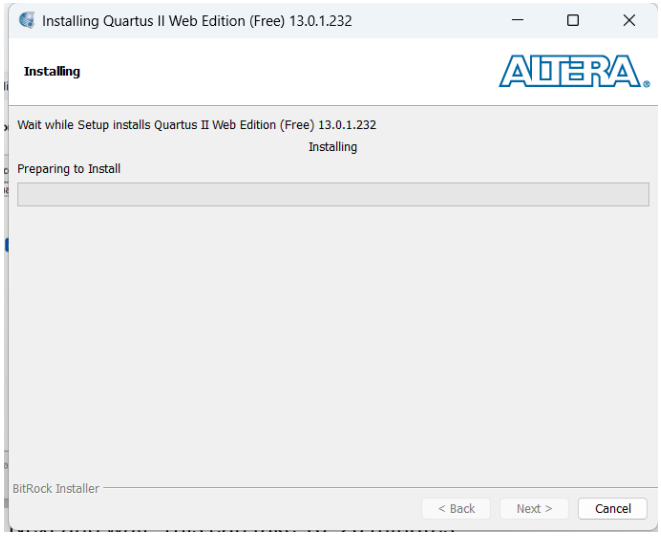
4. **Installation Directory:** Keep it as it is. (Default: C:\altera\13.0sp1)



5. **Select Components:**
    a. Check **Quartus II Software**.
    b. Look under "Devices" and ensure **Cyclone II/III/IV** is checked. If it's greyed out, make sure the .qdz file is in the same folder as the installer.



6. Install: Click Next and wait. This can take 10–20 minutes.

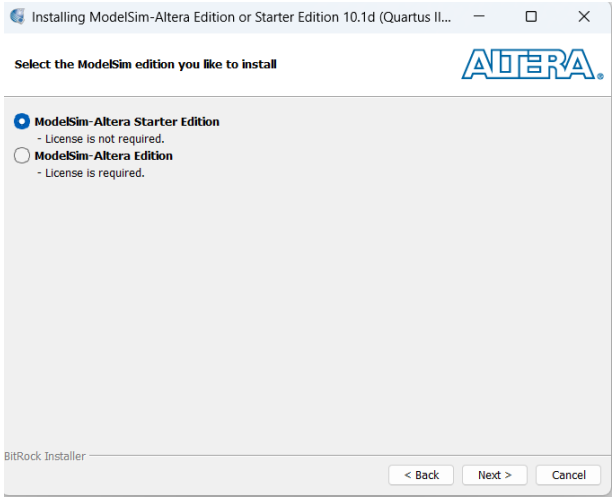


7. Finish: Uncheck "Provide Feedback" and click Finish.

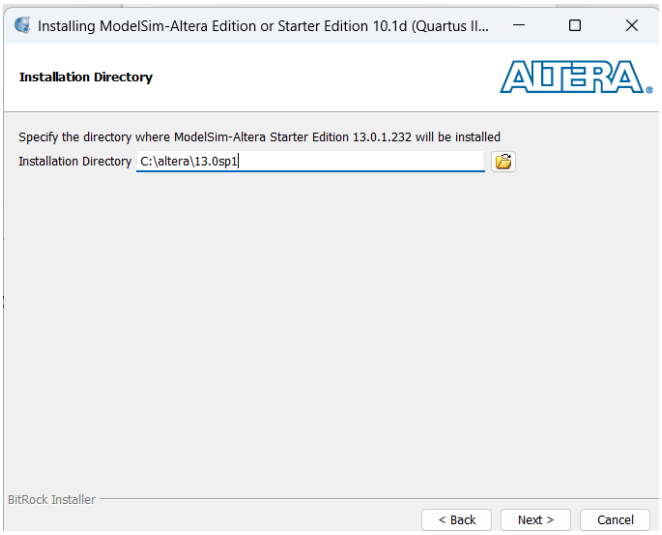
2.1 ModelSim Installation

Once Quartus is finished, go back to the folder where you extracted 1.zip:

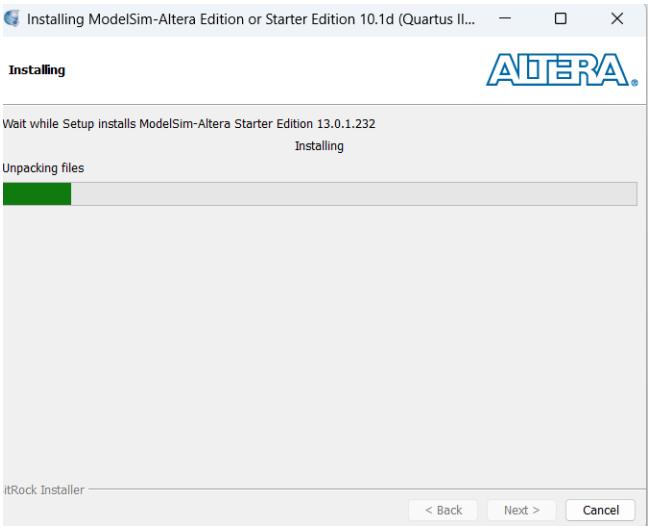
1. Look for a file named ModelSimSetup-13.0.1.232.exe.
2. **Double-click** to run it separately.
3. When asked to choose a version, select ModelSim-Altera Starter Edition (the one that says "Free").



4. **Important:** When it asks for an installation directory, use the same parent folder as Quartus: C:\altera\13.0sp1



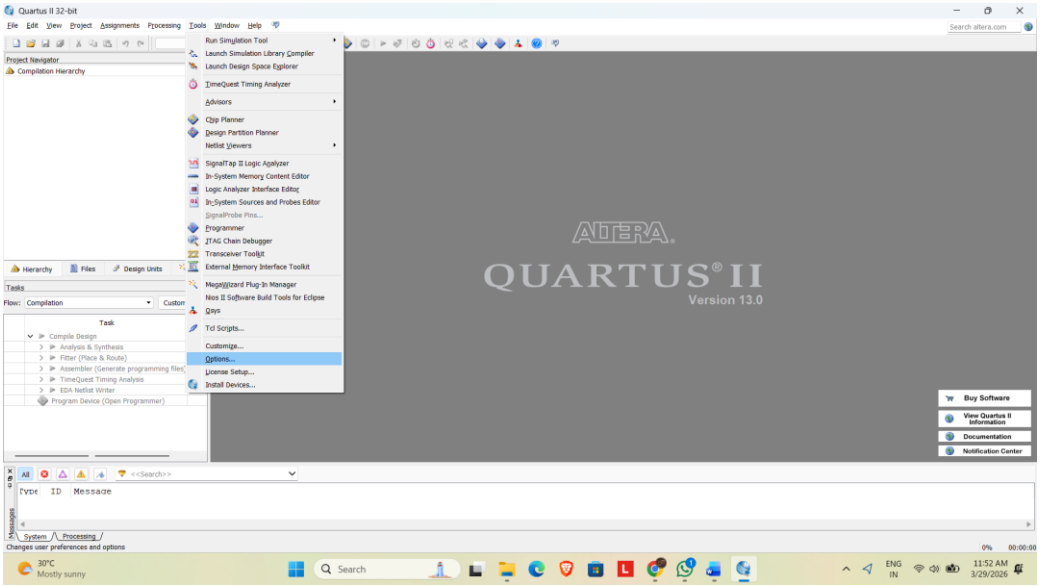
5. Wait till it install



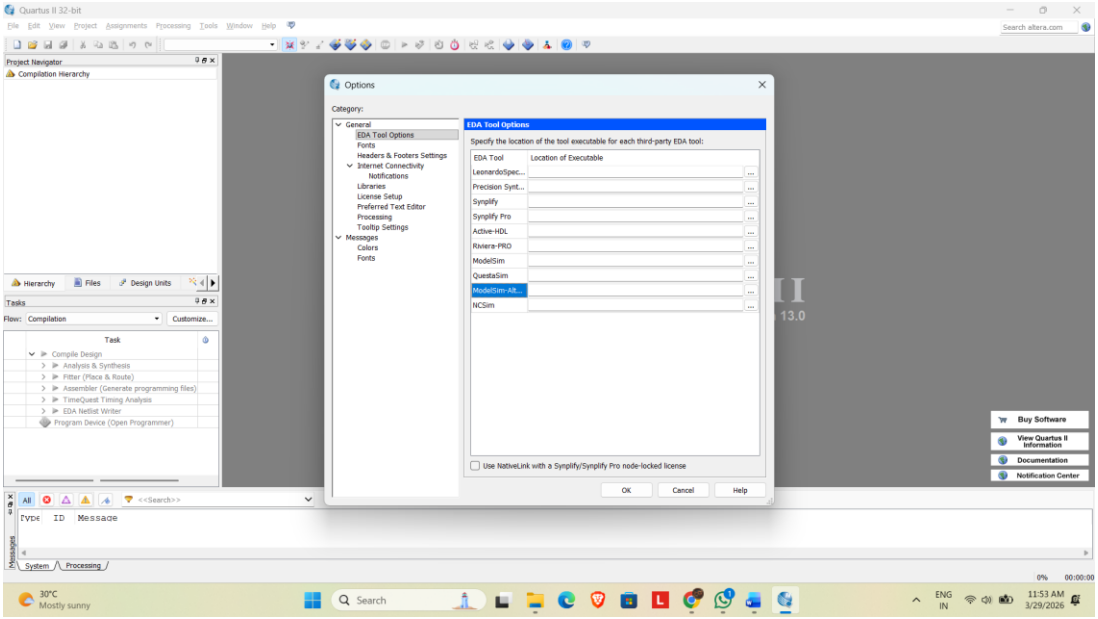
6. Finish: Uncheck "Provide Feedback" and click Finish.

2.2 Linking ModelSim to Quartus

1. Open **Quartus II**.
2. Go to the top menu: **Tools** -> **Options**.

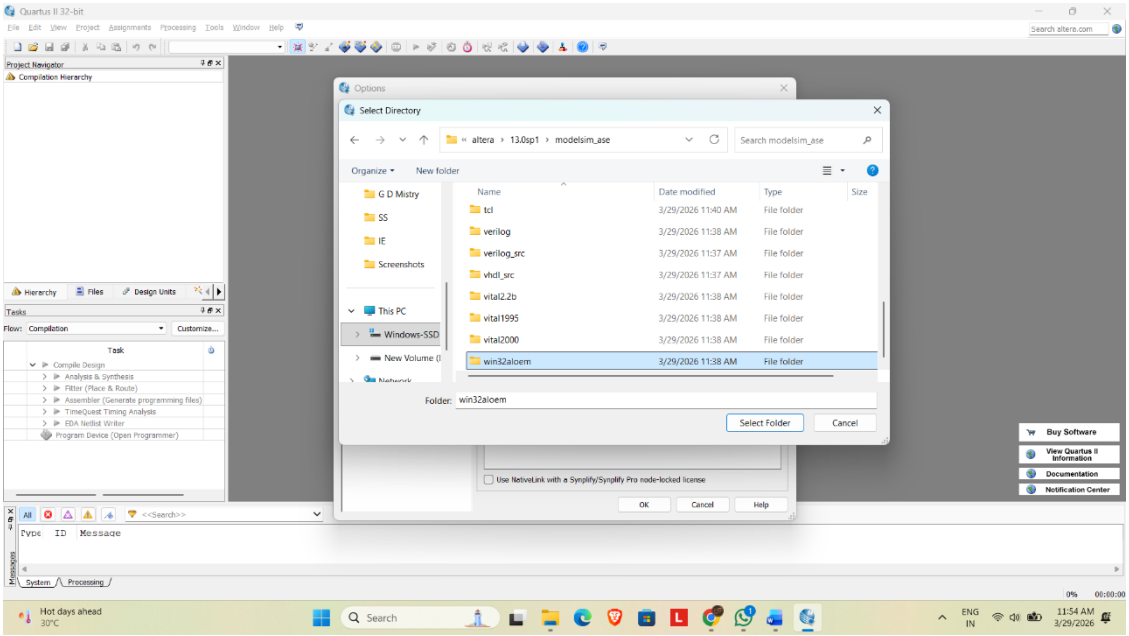


3. On the left sidebar, find and click EDA Tool Options.
4. In the table on the right, look for the row named ModelSim-Altera.

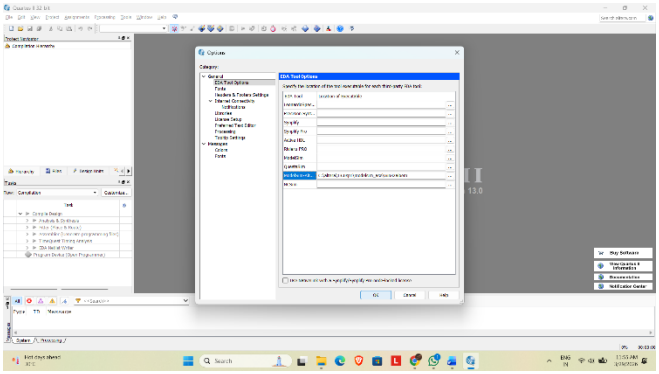


5. Click the "..." button to browse and select this exact folder:

   C:\altera\13.0sp1\modelsim_ase\win32aloem



6. Click **OK**. We have successfully linked ModelSim and Quartus 2 software.

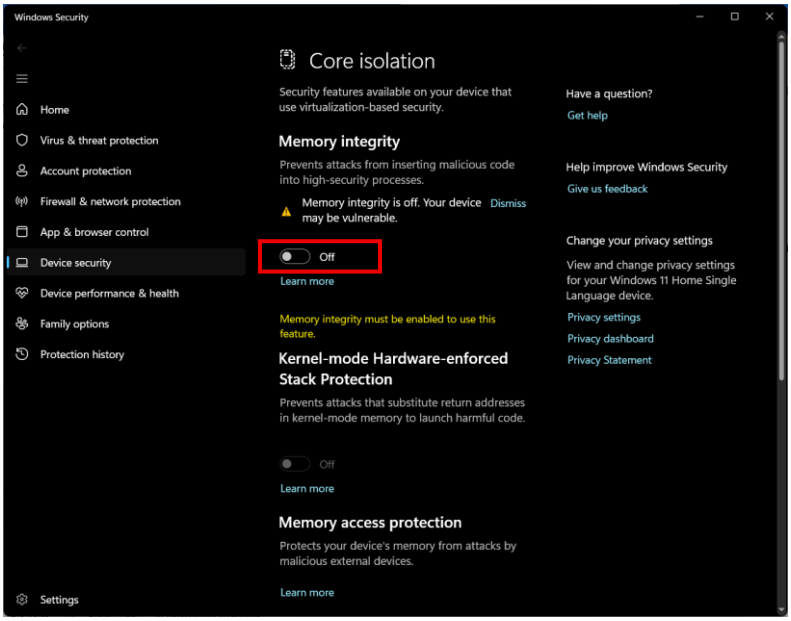


3. Driver Installation

Important: Your computer will not recognise the DE2 board without this driver. Make sure the board is plugged in and powered ON before starting.

Open settings – search “Core isolation “

Make sure it is turn off.



1. Connect & Power the DE2 Board -

    Plug the DE2 board into your PC using the USB cable.

    Press the red power button on the board to switch it ON.

    Wait a few seconds - Windows will attempt (and likely fail) to install a driver automatically.

2. Open Device Manager:

    Press Win + X and choose Device Manager from the list.

    Find the Unknown Device

    In Device Manager, look for one of the following entries (usually with a yellow warning icon):

    . Unknown Device (under Other Devices)

    . USB-Blaster (under Other Devices)

Right-click the entry and select Update Driver.

Browse for Driver Manually

In the Update Driver window, choose:

. Browse my computer for drivers

Click Browse and navigate to the following path:

“C:\altera\13.0spl\quartus\drivers\usb-blaster”

Make sure Include subfolders is checked, then click Next.

3. Accept the Security Warning

If Windows shows a warning popup saying the driver is unsigned, click: · "Install this driver software anyway"

This is safe - the driver is from Altera (now Intel).

4. Restart your laptop

Verify Successful Installation

Once installed, check Device Manager again.

The device should now appear under USB Controllers as: Altera USB-Blaster

The yellow warning icon will be gone.